A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 34

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** 2020 has higher peak flow in the wet season (32 vs 22 $m^3 s^{-1}$ in March) **1** 2020 has much lower dry-season flow (2 vs 9 $m^3 s^{-1}$ in July / August) **1** Flow is much more variable / seasonal in 2020 – the river almost dries up **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** In 1990 snow / ice stored wet-season precipitation and released it gradually as meltwater during the dry season, evening out the flow **1** With less ice, precipitation now falls as rain / melts immediately and runs off quickly, increasing wet-season flow and flood risk **1** Little stored ice remains to melt in the dry season so flow falls very low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3

Do not write outside the box

Question	Marking guidance	Marks
01.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Water shortages for irrigation / drinking / HEP in the dry season **1** Flooding in the wet season **1** Sudden floods from ice-lake outbursts (glacial lake outburst floods) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surface meltwater flows down cracks to the base, lubricating the ice so it slides faster – if this exceeds the faster melting at the front, the front advances **1** But if snowfall at the source has not increased, total volume falls – both retreating and extending glaciers can indicate warming **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1 Gravimetry:** variations in gravity caused by differences in the density and mass of the rock below – ore-bearing rocks are usually denser than the surrounding sedimentary deposits **1 Magnetometry:** rocks that are more magnetic than their surroundings, such as the iron ore magnetite and the ores of tungsten and cobalt **1 Seismic surveys:** echoes of sound waves from controlled explosions or a surface vibrator, giving the depth, density and shape of the rock strata **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Trial drilling is the most expensive technique per sampling site, so the remote surveys are used first to narrow the search to the most promising ground **1** Drilling is the only technique that brings up actual samples of the rock, so it is used to confirm the composition and purity found by chemical analysis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **remote sensing** → collecting data from aircraft or satellites	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reserves are the deposits that can be extracted economically with current technology – the resource base is everything that exists, including deposits that are too low-grade, too deep or too remote to pay **1** A higher price lowers the cut-off ore grade, so deposits that previously could not be mined at a profit become worth extracting and are reclassified as reserves **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.4	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** All the rock above the ore (the overburden) must be stripped off, so habitat is destroyed over a much larger area than with deep mining **1** Spoil heaps of waste rock occupy land, may be unstable and can be colonised only slowly because they hold few nutrients and little water **1** Dust from blasting, crushing and haulage raises atmospheric particulate levels and settles on surrounding vegetation **1** Noise and vibration from blasting and heavy machinery disturbs wildlife and nearby residents **1** Drainage from the workings and spoil is often acidic and carries dissolved metals, polluting watercourses and groundwater **1** Lowering the water table to keep the pit dry dries out nearby wetlands and wells **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
02.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Topsoil and subsoil stripped at the start are stored separately and replaced in the correct order, so the restored soil has a fertile surface layer **1** The land is re-contoured and spoil heaps are given shallow, stable slopes before being covered with soil and planted **1** The site can be returned to agriculture or forestry, or flooded workings developed as a wetland or water-sport habitat **1** Planning restoration first means the topsoil is saved rather than buried, the final landform is built as mining proceeds, and the cost is included in the economic assessment of the mine **1** A restoration bond or condition in the planning permission makes certain the work is paid for even if the company fails **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 136-142; AQA spec 3.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Clay particles are very small and carry electrostatic charges on their surfaces so they repel each other and stay suspended **1** Flocculants (aluminium sulfate / alum or polyelectrolytes) are added and mixed to neutralise the charges **1** The particles clump together (flocs) and settle in the clarifier tank **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Increases dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from anaerobic sources **1** Oxidises some dissolved metals (eg iron) so they become insoluble and can be removed – improves taste **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Peaty water contains organic matter which reacts with chlorine to produce toxic substances (eg trihalomethanes) **1** Ozone / UV kill pathogens without forming these by-products (though chlorine / chloramine keeps water sterile during distribution) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.4	**1 mark each** 1 mark for each of the 3 points below. **1 Ion exchange:** toxic ions adsorb onto polymer resin beads or zeolite (also used to soften water by removing calcium / magnesium) **1** Reverse osmosis / distillation (desalination) - expensive, energy-intensive **1 Activated carbon:** organic molecules causing odour / taste and pesticides adsorb onto the high-surface-area carbon particles **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-128; AQA spec 3.2.2.4.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 180 / 1.2 = 150 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1
04.2	1 mark for the point below. **1** Cadmium and zinc occur together in the same ores so mine drainage / smelting releases both **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1
04.3	**1 mark each** 1 mark for each of the 4 points below. **1** Mine drainage water contaminated irrigation water / paddy soil so rice absorbed cadmium (1.2 vs 0.05 mg kg^{-1}) **1** Cadmium is liposoluble / not excreted so it bioaccumulates in the body (kidneys) with chronic exposure to small doses over years **1** Cadmium causes kidney failure and bone decalcification / skeletal collapse (itai-itai disease) **1** Rice is the staple food so residents receive a large dose – village B has no exposure and no disease (correlation supports causation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **correlation** → a relationship in which the two variables change together • **causation** → one variable actually causing the change in the other **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4

Do not write outside the box

Question	Marking guidance	Marks
04.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Collect mine drainage water and neutralise / raise pH with lime so cadmium precipitates and stays insoluble **1** Cover / cap spoil heaps to prevent leaching and dust **1** Recycle nickel-cadmium batteries / hazardous waste landfill **1** Phytoremediation of contaminated paddy soils **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
04.5	**1 mark each** 1 mark for each of the 2 points below. **1** Manufacture may release cadmium in effluents / dust **1** Panels have a limited life – disposal of old panels (landfill, incineration of pigments) could release cadmium unless recycled **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery **Do not credit** • 'solar panels do not work in cold countries' - output depends on insolation, and cells are slightly more efficient when cool • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
05.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 15 - (-18) = 33 °C **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (°C) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	1

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 4 points below. **1** Visible light passes easily through the atmosphere and is absorbed by the surface, which warms **1** The warm surface emits (long-wave / far) infrared radiation **1** Greenhouse gases (water vapour, CO_2, methane) absorb infrared better than the average for the atmosphere and convert it to heat **1** The warm atmosphere emits infrared which is absorbed by the surface, and it reduces heat loss from land and oceans by conduction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **long-wave** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	4
05.3	1 mark for the point below. **1** Any gas that is better at absorbing infrared radiation than the average for the atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	1
05.4	**1 mark each** 1 mark for each of the 2 points below. **1 Arriving:** mainly ultraviolet, visible light and near infrared (short wavelength) – leaving: mainly long-wavelength (far) infrared **1** Gases that are transparent to the incoming short wavelengths can absorb the outgoing long wavelengths, so changing their concentration changes how much energy is retained **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
05.5	**1 mark each** 1 mark for each of the 2 points below. **1** The energy and the processes it drives control climate, ocean currents and the hydrological cycle **1** These determine the abiotic conditions (temperature, rainfall) that each species is adapted to, so a change alters where species can survive **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** A waste is assumed to cause pollution until research shows it is unlikely to do so **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution **1** If polluters can escape their responsibilities / penalties are low, there is little incentive to comply **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
06.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Marine discharges where currents dilute and disperse emissions **1** Emissions downwind of urban areas **1** Not discharging waste onto permeable rock above an aquifer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeable** → how easily water passes through / lets water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (60 x 45) / 9 **1** = 300 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (beetles) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
07.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Marked individuals mix freely / randomly with the rest of the population **1** No births, deaths, immigration or emigration between samples **1** Marking does not affect survival or behaviour (eg make them more visible to predators) **1** Marks do not wear off / fall off **1** All individuals have the same chance of being caught (traps do not attract or repel marked animals) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Lincoln Index** → estimating population size from the proportion of marked animals recaptured **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	4
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Fewer marked beetles would survive to be recaptured **1** So the proportion marked in the second sample would be too low and the population would be overestimated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Enough time for marked beetles to mix with the population **1** Not so long that births / deaths / migration or loss of marks change the population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 401-402; AQA spec 3.7.2.1.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Low liposolubility and low persistence **Also accept** • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 2 points below. **1 Bioaccumulation:** build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown **1 Biomagnification:** increase in concentration of the pollutant at successive trophic levels of a food chain **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **bioaccumulation** → a pollutant building up inside one organism over time • **biomagnification** → concentration rising at each step along a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** They are intended to be toxic (to the target species) **1** They are deliberately released / spread into the environment (rather than being an accidental or waste release) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
09.1	**1 mark each** 1 mark for each of the 4 points below. **1** Affluent people buy more consumer goods / trivia / disposable items and throw items away rather than repairing them **1 Built-in obsolescence:** products designed to have short lives / wear out / go out of fashion so customers upgrade **1** Disposable products (razors, lighters, pens) that must be replaced **1** Over-packaging to make products attractive **OR** increases wastes from manufacture and mineral extraction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
09.2	**Any 3 of 7** 1 mark each for any 3 of the 7 points below, in any order, maximum 3. **1 Properties of the waste:** toxicity, flammability, degradability, radioactivity **1** Population density (collection costs) **1** Land availability for landfill **1** Availability of recycling technology / markets **1** Legislation / regulatory framework (landfill tax) **1** Public willingness to recycle / environmental awareness **1** Processing costs / household income **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1 Advantage:** aerobic decomposition produces little methane – nutrients / organic matter returned to soil (circular economy) – reduces landfill volume **1 Disadvantage:** needs separation / collection of clean organic waste – odour / pests / pathogens if poorly managed – low-value product – not suitable for mixed waste **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
09.4	1 mark for the point below. **1** Many different materials are mixed together / contaminated so identification and separation are costly and the quality of recovered material is reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-282; AQA spec 3.4.3.2.12.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Carbon:** combustion of fossil fuels and biomass moving carbon from long-term stores to the atmosphere – deforestation reducing photosynthesis – ploughing / drainage increasing aerobic decomposition – new anaerobic environments (paddies, landfill, reservoirs, livestock) releasing methane – marine pollution reducing phytoplankton – consequences: rising CO2 and CH4, climate change, ocean acidification, reduced soil organic matter • **Nitrogen:** Haber process doubling fixation – legume cultivation – drainage increasing nitrification and reducing denitrification – combustion NOx – sewage and fertiliser leaching – consequences: eutrophication, smogs, acid rain, N2O warming, NOx toxicity • **Counter-strategies carbon:** renewables / nuclear replacing fossil fuels, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, increasing soil organic matter (low tillage, mulching), CCS (pre / post combustion, storage) • **Counter-strategies nitrogen:** reduced combustion, catalytic converters / urea sprays, biological fixation instead of Haber, legumes and crop rotation, management of biological wastes as fertiliser, low-solubility fertilisers, timing, buffer strips, low tillage, nitrate vulnerable zones • **Evaluation:** some strategies well established and cheap (rotation, buffer strips), others expensive or unproven (CCS) – scale of fossil fuel dependence – feedback and time lags – link to circular economy and low-temperature natural processes – judgement	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **nitrification** → ammonium oxidised to nitrite and then nitrate • **denitrification** → nitrate converted back to nitrogen gas • **leaching** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	
	Model answer - Level 3 (8 marks)	
	Humans have shifted the carbon cycle by burning fossil fuels and biomass, moving carbon from long-term stores into the atmosphere faster than photosynthesis and ocean uptake remove it, while deforestation reduces photosynthesis, ploughing and drainage speed aerobic decomposition, and paddies, landfills and livestock create new anaerobic sources of methane. The nitrogen cycle has changed as much: the Haber process has roughly doubled fixation, legume crops add more, drainage increases nitrification and reduces denitrification, combustion produces NOx, and leached fertiliser and sewage nitrate cause eutrophication, while nitrous oxide adds to warming.	
	Carbon can be counteracted by replacing fossil fuels with renewables and nuclear power, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, and increasing soil organic matter through low tillage and mulching, with carbon capture and storage as an unproven option. For nitrogen, catalytic converters and urea sprays reduce combustion NOx, biological fixation by legumes and rotation can replace manufactured nitrate, biological wastes can be managed as fertiliser, and low-solubility fertilisers, careful timing and buffer strips reduce leaching.	
	Some strategies are cheap and established - rotation, buffer strips, fertiliser timing - but others are expensive or untested, and the scale of fossil-fuel dependence, together with feedback and time lags, means the atmosphere will respond slowly. The strategies that mimic natural low-temperature cycling, as in the circular economy, are the most sustainable; the most effective is simply to leave carbon in the ground.	
	Weaker answer - Level 1 (3 marks)	
	Humans have changed the carbon cycle by burning fossil fuels which releases carbon dioxide, and by cutting down trees. They have changed the nitrogen cycle by using fertilisers which run off into rivers and cause eutrophication. To counteract this we could use renewable energy instead of fossil fuels and plant more trees. We could use less fertiliser. These would help the cycles go back to normal.	
	What makes the difference	
	• Level 1 names one or two changes per cycle; Level 3 covers several processes for each (decomposition, methane sources, Haber process, nitrification, NOx).	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 connects each counter-strategy to the process it reverses; Level 1's strategies are general.	
	• The evaluation in Level 3 distinguishes established from unproven strategies and comments on scale and time lags - Level 1 has no evaluation.	
10.2	1 mark for the point below. **1** The average time an atom / molecule remains in the reservoir before being transferred out (volume / transfer rate) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-157; AQA spec 3.2.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Nature of the problem:** interconnected atmosphere-hydrosphere-biosphere systems – changes over different time and spatial scales – natural fluctuations (solar output, orbit) masking or exaggerating anthropogenic change – time delay between cause and effect (ocean heat capacity) • **Limitations of historic data:** sparse global coverage, unsophisticated equipment, heat island bias, no atmospheric composition records • **Proxy data and their reliability:** dendrochronology, coral growth rings, pollen in dated sediments, ice cores (gas bubbles, isotope ratios, volcanic ash layers, compaction problems) • **Modern monitoring:** long-term data sets (Mauna Loa), electronic equipment, satellites (LEO vs geostationary), Argo floats, remote sensing of ice thickness / area / sea level • Feedback mechanisms and tipping points adding uncertainty – ecological uncertainty (species survival, distribution, fragmentation) • **Computer models:** how they are tested and refined – importance of accurate representative data – scenario-based predictions and ranges	25
	• Uncertainty over ENSO / Conveyor / jet stream responses • **Evaluation and conclusion:** confidence in the overall trend and human cause despite detailed uncertainty – role of peer review and international bodies (IPCC) *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90, 100-104, 10; AQA spec 3.2.1.2, 3.1.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • **Fossil fuels:** choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports – new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • **Nuclear:** high energy density reduces transport – waste properties (half-life) determine storage – thorium and fusion prospects • **Renewables:** matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting) – intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity – low-impact designs (helical turbines, VAWT, tidal lagoons) • **Energy conservation reduces demand:** transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use	25
	– life-cycle / embodied energy • **Evaluation:** no resource is impact-free – knowledge allows the best mix and mitigation – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

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